

OSCEOLA, WI, USA

WMO: 726418

Lat: 45.308N

Lon: 92.690W

Elev: 903

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54912

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-8.8	-20.8	1.8	-11.7	-17.0	2.2	-7.9	20.9	19.6	19.3	19.1	2.6	290	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	88.5	73.8	85.7	72.3	82.4	70.0	77.6	84.9	75.1	82.2	73.0	79.8	10.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	136.3	81.7	72.8	125.7	79.0	71.0	118.1	77.4	41.5	84.7	39.1	82.8	37.2	80.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
20.5	18.3	16.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-21.6	93.9	5.5	3.2	-25.6	96.2	-28.7	98.0	-31.8	99.8	-35.8	102.1
			-22.0	80.3	5.4	2.7	-25.9	82.2	-29.0	83.8	-32.1	85.3	-36.0	87.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	15.0	17.5	30.3	45.2	57.2	66.7	71.0	67.7	60.9	47.3	34.2	20.1	
	DBStd	22.01	12.83	12.41	11.91	10.01	8.57	6.56	5.46	5.98	8.47	9.12	10.38	12.17	
	HDD50	4428	1085	910	615	206	27	0	0	1	11	162	483	928	
	HDD65	7974	1550	1330	1075	597	270	56	12	38	176	553	924	1393	
	CDD50	2448	0	0	5	62	251	502	652	549	338	80	9	0	
	CDD65	516	0	0	0	3	28	108	198	121	53	5	0	0	
	CDH74	4734	0	0	1	45	335	977	1846	1047	442	41	0	0	
	CDH80	1325	0	0	0	10	99	272	561	274	105	4	0	0	
Wind	WSAvg	6.8	6.8	7.0	7.1	8.2	7.8	6.6	5.4	5.0	6.3	6.9	7.2	6.8	
Precipitation	PrecAvg	32.8	0.9	0.8	1.7	2.8	3.9	4.6	4.3	4.0	3.7	3.1	1.8	1.1	
	PrecMax	43.6	2.2	1.8	3.2	8.1	7.8	7.7	7.8	8.5	8.6	7.1	4.7	2.5	
	PrecMin	23.8	0.2	0.1	0.6	0.7	1.2	1.9	2.2	1.0	1.1	1.0	0.0	0.2	
	PrecStd	5.3	0.6	0.4	0.7	1.6	1.9	1.5	1.7	1.9	1.9	1.9	1.4	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	50.8	66.1	80.8	89.5	91.0	92.7	90.5	88.0	79.4	66.1	47.0	
		MCWB	38.0	45.2	53.7	62.8	68.0	75.3	77.1	75.1	72.1	62.6	55.6	44.1	
	2%	DB	38.3	43.3	58.3	72.4	81.9	86.3	89.1	86.3	82.5	72.9	60.5	42.8	
		MCWB	35.0	38.4	50.2	55.6	65.7	70.7	75.7	73.5	70.0	61.0	52.1	39.0	
	5%	DB	35.8	38.6	52.2	67.4	77.2	82.3	85.8	82.3	79.3	67.6	54.4	37.6	
		MCWB	33.9	34.8	46.3	52.7	62.2	68.9	73.5	70.8	67.6	58.1	47.5	35.2	
	10%	DB	32.4	35.6	46.1	63.2	72.7	80.2	82.4	80.6	75.2	63.0	49.7	35.9	
		MCWB	30.9	32.7	41.0	51.0	59.9	67.6	71.0	69.6	65.7	55.0	44.4	34.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	45.3	56.8	64.4	72.7	80.7	82.3	79.2	76.0	68.9	58.7	46.0
			MCDB	42.8	50.8	61.5	76.0	84.3	84.7	86.9	85.5	82.3	73.3	63.7	46.4
2%		WB	35.6	38.9	51.1	58.9	68.7	75.3	78.3	75.7	73.0	62.7	53.1	39.2	
		MCDB	37.7	42.7	56.9	68.9	76.8	82.3	85.2	82.8	79.5	68.8	58.9	41.5	
5%		WB	33.6	35.2	46.1	54.9	65.9	72.2	75.6	73.5	70.3	59.4	48.0	36.3	
		MCDB	35.5	38.1	51.3	64.0	72.8	79.3	83.1	79.7	76.6	66.4	52.8	37.9	
10%		WB	30.9	32.4	41.6	52.0	63.1	69.6	73.4	71.3	67.6	55.7	44.6	34.1	
		MCDB	32.3	34.9	46.3	61.4	69.2	76.0	80.4	77.3	73.9	61.8	49.3	35.6	
Mean Daily Temperature Range	5% DB	MDBR	16.6	18.4	18.0	20.1	21.4	19.8	20.5	20.7	21.0	18.9	15.8	14.4	
		MCDBR	16.8	18.8	23.5	28.5	27.4	23.2	23.2	23.3	23.7	25.7	22.7	15.6	
		MCWBR	13.9	14.6	15.3	15.7	13.1	11.3	11.7	12.0	12.5	14.9	15.1	13.3	
	5% WB	MCDBR	16.0	17.3	22.2	24.6	21.6	19.5	19.5	19.9	19.9	20.6	20.0	13.8	
		MCWBR	13.7	14.0	15.2	15.4	12.5	12.3	11.6	12.0	12.1	13.9	14.5	12.0	
Clear-Sky Solar Irradiance	taub	0.259	0.279	0.319	0.371	0.403	0.392	0.385	0.397	0.368	0.319	0.288	0.265		
	taud	2.401	2.337	2.348	2.303	2.259	2.328	2.373	2.348	2.401	2.526	2.509	2.421		
	Ebn at Noon	275	289	290	282	275	278	278	270	269	272	262	260		
	Edh at Noon	23	29	34	39	42	39	37	37	32	24	21	20		
All-Sky Solar Radiation	RadAvg	542	887	1228	1467	1690	1875	1983	1654	1284	807	512	377		
	RadStd	60	86	107	141	137	141	67	120	88	108	46	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	-1.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page